

Sample OULAD VLE Fairness Audit Report

Learning Analytics Demonstration

AI system assessed

Field	Value
System type	VLE engagement analytics / early-alert style monitoring
Deployment category	Current or near-future deployment risk
Student impact area	Learning support and intervention
Risk tier	Moderate to high depending on use
Primary concern	Behavioural proxy bias and unequal flagging
Required control	Human review before intervention or escalation

Why VLE analytics needs fairness auditing

VLE analytics can appear neutral because it uses behavioural indicators such as clicks, logins, submission timing, and activity records.

However, engagement data may encode structural differences.

Example risks:

- Caring responsibilities may affect login timing.
- Employment responsibilities may affect engagement frequency.
- Subject area may shape expected VLE activity.
- Socioeconomic conditions may affect access and consistency.
- Platform behaviour may not reflect actual learning effort.

The audit question is:

t; Are VLE-based indicators creating unequal support, suspicion, or intervention patterns across student groups?

Dataset preparation summaryThe prepared OULAD student feature table should include:

Field	Purpose
Gender	Subgroup fairness analysis
Region	Potential proxy-risk variable

Highest education	Background and proxy-risk analysis
IMD band	Socioeconomic proxy-risk analysis
Disability	Future intersectional fairness analysis
Total VLE clicks	Engagement feature
Assessment score average	Outcome indicator
Final result	Course outcome
Low engagement flag	Demonstration flag for audit testing

Metrics to review

Metric	Why it matters
Representation by gender	Checks subgroup sample size
Average VLE clicks by gender	Shows engagement distribution differences
Low engagement flag rate by gender	Tests unequal flagging risk
Unsuccessful outcome rate by gender	Identifies outcome gaps
Proxy association with gender	Identifies variables requiring governance review

Proxy-bias review

Potential proxy variables in OULAD include:

- Region
- IMD band
- Highest education
- Age band
- Studied credits
- Previous attempts

- Disability
- Subject module

A high proxy association score does not prove bias. It means the variable may indirectly encode a protected or sensitive characteristic and should be reviewed before use in decision-making.

Example governance decision

Recommendation Rationale

Approve with monitoring VLE analytics may support students, but indicators should remain advisory and must be monitored for unequal flagging or intervention patterns

Required mitigations

1. Do not treat VLE engagement as a direct measure of student ability.
2. Require human review before labelling a student as at-risk.
3. Monitor low-engagement flag rates across gender and other groups.
4. Review proxy variables before modelling.
5. Explain to students how engagement data is used.
6. Provide correction routes where data is inaccurate.
7. Reassess fairness after each academic cycle.